1. Password-Cracking Countermeasures

- **Strong Password Policies:** Enforce complex, long passwords (mix of uppercase, lowercase, numbers, symbols).
- **Account Lockout Mechanisms:** Lock accounts after multiple failed login attempts to prevent brute force or online guessing.
- **Multi-Factor Authentication (MFA):** Requires additional verification (e.g., OTP, biometrics).
- **Salting and Hashing Passwords:** Add unique random data (“salt”) before hashing passwords to prevent precomputed rainbow table attacks.
- **Use of Slow Hash Functions:** Algorithms like bcrypt, scrypt, or Argon2 that slow down brute force attacks.
- **Monitoring and Alerting:** Detect unusual login attempts or patterns.
- **Limit Password Reuse:** Force periodic changes and disallow old passwords.

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2. Active and Passive Online Attacks

- **Active Online Attacks:** Attacker interacts directly with authentication systems (e.g., brute force, dictionary attacks) trying passwords live.
- **Passive Online Attacks:** Attacker eavesdrops on authentication traffic or uses session hijacking but does not directly try passwords.

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3. Offline Attacks

- Attacker obtains password hashes (e.g., via data breach) and tries to crack them offline using computing power.
- More dangerous because no account lockout or detection.
- Uses tools like John the Ripper, Hashcat, rainbow tables.

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4. Keyloggers and Other Spyware Technologies

- **Keyloggers:** Software or hardware that records keystrokes to steal passwords.
- **Screen scrapers:** Capture screen content.
- **Form grabbers:** Capture data entered into forms.
- Spyware often installed via phishing, malicious downloads, or Trojans.

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5. Trojans and Backdoors

- **Trojans:** Malware disguised as legitimate software that allows attackers unauthorized access.

- **Backdoors:** Hidden access points left by attackers to enter a system later bypassing authentication.

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6. Overt and Covert Channels

- **Overt channels:** Legitimate communication paths (e.g., network protocols, emails).
- **Covert channels:** Hidden communication methods used to exfiltrate data secretly, such as hiding data in packet headers or timing channels.

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7. Types of Trojans

- **Remote Access Trojans (RATs):** Allow remote control of infected system.
- **Downloader Trojans:** Download and install other malware.
- **Spy Trojans:** Spy on user activities.
- **Backdoor Trojans:** Provide unauthorized remote access.
- **Rootkit Trojans:** Hide presence and maintain persistence.

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8. Reverse-Connecting Trojans

- Unlike direct connection Trojans, these initiate a connection from inside the victim's network back to the attacker, bypassing firewall restrictions that block incoming connections.

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9. Netcat Trojan

- **Netcat:** A versatile networking utility used by attackers as a backdoor or Trojan.
- Can listen on ports, create reverse shells, transfer files.
- Often used for remote administration by attackers once inside.

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10. Indications of Trojan Attacks

- Unexpected system slowdowns or crashes.
- Unusual network activity or unknown outbound connections.
- Disabled antivirus or firewall.
- Unknown processes running.
- Changes in system files or settings.

- Presence of suspicious files or services.